A-A-1923
May 2, 1983

COMMERCIAL ITEM DESCRIPTION

SHIELD, EXPANSION
(LAG SHIELD)

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification FF-S-325.

This description covers lag bolt expansion shields for attaching equipment and fixtures to an existing solid masonry.

Salient characteristics:

General

Shall consist of a two piece expansion shield which opens at the bottom and spreads out into a cone when the lag bolt is tightened.

Shall be of zinc alloy or malleable iron.

Shall be supplied by bolt diameter as specified according to table I below and by the following classes, as specified:

Class 1 - Long shield anchors
Class 2 - Short shield anchors

TABLE I

Size, diameter of lag bolt (inches)	Shield Dimensions (inches)			Proof Test Loads (pounds) (minimum)			
		Minimum Length		Malleable Iron Shields		Zinc Alloy Shields	
	Outside diameter minimum	Class 1 long shield anchors	Class 2 short shield anchors	Class 1 long shield anchors	Class 2 short shield anchors	Class 1 long shield anchors	Class 2 short shield anshors
1/4	7/16	1-1/2	1	200	160	500	400
5/16	1/2	1-3/4	1	480	380	1000	800
3/8	5/8	2-1/2	1-3/4	960	640	1600	1300
7/16	5/8	2-3/4	2	1250	950		
1/2	3/4	3	2	1600	1300	2800	2100
5/8	7/8	3-3/8	2	1725	1400		
3/4	1-1/8	3-1/2	2	1850	1500		

Shield body

Shall consist of two semi-cylindrical hollow sections interlocked at the top of ears or lugs.

Bore of the shield shall be tapered and have an internal thread for about two-thirds of its distance from the bottom.

The thread of the shield shall correspond to the thread of the specified lag bolt to be used in the shield (UNC-2B coarse thread series).

The outer walls of the shield sections shall have circular ribs for better gripping and holding action of the shield.

The average thickness of the thinnest wall section shall be not less than 1/32 of an inch.

Test. Shall withstand minimum proof test loads specified in table I when installed with the applicable carbide tipped masonry drill of ANSI B 94.12 and tested in accordance with the tension test of ANSI/ASTM E 488. There shall be no evidence of failure when performing the tension test of -40° and $+170^{\circ}$ F.

Workmanship. The expansion shields shall be free from defects such as rust, scale, fins, or any other defect which may affect their use and serviceability.

The issues of ANSI B 94.12 and ANSI/ASTM E 488 in effect on the date of the solicitation shall be used to determine compliance with these requirements.

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Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Note. Purchaser should specify class and bolt diameter.

ASTM standards are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

ANSI standards are available from the American Standards Institute, Inc., 1430 Broadway, New York, NY 10018.

MILITARY INTERESTS:

Coordinating Activity

Army - AR

Custodians

Army - AR
Navy - YD
Air Force - 99

PREPARING ACTIVITY:

GSA - FSS

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